

Dictionary L3 — Chirality axis: parity violation is a consequence of the substrate being a COMPLEX (Hermitian) domain. The same $SO(2)$ complex structure (the ‘i’) that makes D_{IV}^5 a bounded SYMMETRIC domain forces physics to be chiral.

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Dictionary L3 — Chirality (parity violation from the complex structure)

0. Where this sits + the L2 handle

L2 derived the Region axis (Shilov stabilizer = $SO(4) = SU(2)_L \times SU(2)_R$) and surfaced the chirality handle: parity violation = the boundary breaking the $L \leftrightarrow R$ symmetry of $SO(4)$. L3 derives WHAT breaks it — and the answer is the deepest structural fact about D_{IV}^5 : it is a COMPLEX (Hermitian) domain, and the complex structure is what makes physics chiral.

1. The complex structure IS the $SO(2)$ factor (rigorous)

$D_{IV}^5 = SO_0(5,2)/[SO(5) \times SO(2)]$ is a HERMITIAN symmetric domain — a bounded symmetric domain with a complex structure. That complex structure J is exactly the $SO(2)$ factor of the maximal compact $K = SO(5) \times SO(2)$: the $SO(2) = U(1)$ acts as the global complex structure (the substrate’s “i”). This is the same $SO(2)$ that: - gives the ρ -vector its second component ($\rho_2 = N_c/\text{rank}$, the Shilov/Wallach direction), - supplies the $U(1)$ CHARGE (L2), - and now: provides the COMPLEX STRUCTURE.

The holomorphic tangent space at the origin is $T^\wedge(1,0) = (\text{vector of } SO(5)) \otimes (SO(2)\text{-charge } +1) = 5_{+1}$ (complex dim 5); the antiholomorphic is $T^\wedge(0,1) = 5_{-1}$. J acts as $+i$ on $T^\wedge(1,0)$ and $-i$ on $T^\wedge(0,1)$.

2. The Hardy space is holomorphic $\rightarrow$ intrinsically chiral (the mechanism)

The leptons live in the Hardy space $H^2(S)$ over the Shilov boundary (L2). The Hardy space consists of boundary values of HOLOMORPHIC functions — not antiholomorphic. So the lepton sector is intrinsically chiral: it sits in the holomorphic half, distinguished by J .

Chirality (L/R) = holomorphic vs antiholomorphic = the sign under the complex structure J ($SO(2)$).

And the parity violation follows immediately:

Because the substrate is a HOLOMORPHIC domain, the Hardy space picks ONE chirality. The weak $SU(2)_L$ is the $SU(2)$ of the $SO(4)$ stabilizer that is COMPATIBLE with J (acts holomorphically); $SU(2)_R$ is the antiholomorphic one. The holomorphic Hardy space couples to $SU(2)_L$ — left-handed weak coupling. The right-handed (antiholomorphic) sector is NOT in the Hardy space $\rightarrow$ right-handed fermions are weak singlets.

3. The headline: parity violation = the substrate being complex

The deep point: the SAME complex structure (the $SO(2)$, the “i”) that makes D_{IV}^5 a bounded SYMMETRIC domain — that defines it AS a Hermitian domain — is what forces physics to be chiral. They are not two facts; they are one.

- A REAL (non-Hermitian) symmetric domain would have no preferred holomorphic/antiholomorphic split → parity-SYMMETRIC physics.
- D_{IV}^5 is Hermitian (complex) → the holomorphic Hardy space breaks $L \leftrightarrow R$ → maximal parity violation.

So parity violation is a consequence of the substrate being complex. Nature does not waste the “i”: the complex structure that makes the domain a bounded symmetric domain (and gives the charge, and the Shilov direction) ALSO does the work of breaking parity. This is the “simplest mechanism” pattern — one structure, multiple jobs.

4. The 5-tuple after L3

5-tuple axis	status
σ_{BF}	DERIVED (keystone — $SO(5)$ -weight parity)
Region	DERIVED (L2 — Hardy/Shilov vs Bergman/bulk; lepton-on-Shilov + colorlessness)
Chirality	DERIVED-geometric (L3) — holomorphic/antiholomorphic = sign under J ($SO(2)$ complex structure); parity violation = the substrate being Hermitian. Exact L/R spinor assignment pinned-not-asserted ($SO(5,2)$ embedding)
Charge	LOCATED (L2 — $SO(2)$ + GMN, constrained by $\sin^2\theta_W=2/9$)
particle/antiparticle	DERIVED (engine v0.4 — E/F)
Winding/generation	OPEN (gate; #414 lean route II, count over-determined, mechanism open)

The static 5-tuple is now essentially DERIVED: σ_{BF} , Region, Chirality (geometric), particle/antiparticle — with Charge located (constrained) and only the generation/winding axis open (the deepest gate, #414). Four of the five static structural axes are derived from the geometry of D_{IV}^5 ; the fifth (charge) is located and constrained; the dynamic axis (generation) is the open gate.

5. Honest scope + tier

RIGOROUS (geometry): D_{IV}^5 is Hermitian symmetric; J = the $SO(2)$ factor; $T^{\wedge}(1,0) = 5_{+1}$, $T^{\wedge}(0,1) = 5_{-1}$; the Hardy space = holomorphic boundary values; $SU(2)_L/SU(2)_R = J$ -compatible/antiholomorphic halves of the $SO(4)$ stabilizer.

DERIVED-geometric (this doc, modulo the keystone bet): chirality = holomorphic/antiholomorphic (sign under J); parity violation = the substrate being a holomorphic domain (the Hardy space picks one chirality). This is a structural mechanism, not an assignment.

PINNED-not-asserted: the exact L/R Weyl-spinor assignment (which 2 of the Dirac 4 is left-handed) needs the $SO(5,2) \rightarrow$ Lorentz embedding — flagged in L2, still flagged. The MECHANISM (chirality from J, parity violation from holomorphy) is robust without it; the exact labeling is the pin.

Cal #27 / honesty: I’m claiming the MECHANISM of chirality/parity-violation (the complex structure → holomorphic Hardy space → one chirality) is geometric and robust — and it’s genuinely deep (parity violation = the substrate being complex). I’m NOT asserting the exact L/R spinor labeling (needs the embedding). And it rides on the keystone bet (K-types = particles) like the rest of the dictionary. The headline (“parity violation

= the substrate being Hermitian”) is a mechanism claim, well-founded; the per-component labeling is pinned downstream.

Next (my lane): with the static 5-tuple essentially derived, move to L4 (masses-from-the-engine) — the radial/Casimir structure giving mass scales — and the #416 per-particle dictionary layer (flip per-cell 5-tuples for Grace’s table). Charge GMN-coefficient pin (against $\sin^2\theta_W=2/9$) when an external check is available.

— Lyra, Dictionary L3 v0.1 (Chirality). GEOMETRIC RESULT: D_{IV}^5 is a HERMITIAN symmetric domain; its complex structure J = the $SO(2)$ factor of K . The Hardy space $H^2(S)$ on the Shilov boundary = boundary values of HOLOMORPHIC functions $\rightarrow$ intrinsically chiral. Chirality (L/R) = holomorphic/antiholomorphic = sign under J . PARITY VIOLATION IS BUILT IN: the substrate is a holomorphic domain, so the Hardy space picks ONE chirality ($SU(2)_L$ = the J -compatible $SU(2)$ of the $SO(4)$ stabilizer; $SU(2)_R$ antiholomorphic = weak singlet). The headline: the SAME complex structure ($i = SO(2)$) that makes D_{IV}^5 a bounded SYMMETRIC domain FORCES physics to be chiral — parity violation is a consequence of the substrate being COMPLEX. A real domain would be parity-symmetric. Exact L/R spinor assignment pinned-not-asserted ($SO(5,2)$ embedding). The static 5-tuple (σ_{BF} , Region, Chirality, particle/antiparticle) is now essentially DERIVED; Charge located+constrained; only generation/winding open.